

URMIA INTL, Iran, Islamic Republic of

WMO: 407120

Lat: 37.6581N

Lon: 45.0560E

Elev: 4344

StdP: 12.53

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.6	15.4	2.9	7.5	13.5	6.7	9.1	18.5	19.2	38.3	16.6	37.8	5.7	270	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.3	93.4	63.6	91.1	63.5	88.2	63.1	67.5	86.1	66.3	85.0	65.1	83.8	6.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	94.6	76.9	59.2	88.5	76.5	57.6	83.2	75.5	34.6	86.0	33.5	85.2	32.5	84.0	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	15.2	12.2	8.1	97.2	5.1	3.2	4.4	99.5	1.4	101.4	-1.5	103.2	-5.2	105.6
			6.7	71.2	4.9	2.0	3.2	72.6	0.3	73.8	-2.5	75.0	-6.1	76.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.5	30.0	34.6	43.4	52.2	61.0	69.9	75.4	74.7	66.5	55.9	43.4	33.7	
	DBStd	16.64	6.57	6.79	6.75	5.98	5.05	4.44	3.75	3.52	4.69	5.53	6.03	7.11	
	HDD50	2046	619	432	224	42	0	0	0	0	0	14	208	507	
	HDD65	5085	1084	852	670	385	141	12	0	0	37	284	649	971	
	CDD50	3320	0	0	20	108	342	596	788	767	494	197	8	0	
	CDD65	887	0	0	0	1	18	158	324	302	82	2	0	0	
	CDH74	10807	0	0	1	33	295	1914	3668	3545	1269	82	0	0	
	CDH80	4335	0	0	0	4	32	632	1673	1627	363	4	0	0	
Wind	WSAvg	5.7	4.7	5.4	6.7	6.7	6.4	6.2	5.8	5.6	5.6	5.4	4.9	4.5	
Precipitation	PrecAvg	10.7	1.1	0.8	1.5	2.0	1.3	0.2	0.3	0.1	0.2	0.8	1.4	0.9	
	PrecMax	19.1	3.1	2.3	3.2	4.6	4.5	1.0	1.6	0.9	1.1	5.2	5.1	3.1	
	PrecMin	6.6	0.0	0.0	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	PrecStd	3.0	0.7	0.6	0.9	1.3	1.1	0.3	0.4	0.2	0.3	1.3	1.1	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.0	57.4	71.4	78.4	84.0	91.6	97.2	96.9	91.3	80.2	65.1	57.2	
		MCWB	40.1	44.3	50.0	55.8	58.6	61.8	64.6	63.3	61.2	56.5	48.9	44.5	
	2%	DB	46.7	53.3	64.5	72.5	80.3	89.2	94.6	93.4	87.4	76.4	61.2	51.7	
		MCWB	38.4	41.7	47.9	52.8	57.8	62.0	64.5	63.4	60.1	55.3	47.2	41.9	
	5%	DB	43.9	49.7	60.5	68.3	77.2	86.2	91.2	91.1	84.1	73.1	58.2	48.0	
		MCWB	36.9	40.0	45.8	51.4	57.3	61.5	64.4	63.6	59.5	54.3	46.2	40.0	
	10%	DB	40.7	46.4	56.8	64.8	74.6	83.9	88.2	88.2	81.0	69.7	55.1	44.5	
		MCWB	35.0	38.3	43.8	50.0	56.3	61.1	64.1	63.3	58.8	53.2	45.2	38.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.3	45.6	51.3	57.5	62.9	67.2	70.9	68.6	64.8	58.5	52.0	45.5
			MCDB	48.2	55.7	67.7	73.2	76.6	84.4	87.4	86.9	83.4	73.8	58.6	54.6
2%		WB	39.4	42.6	48.6	54.7	60.4	65.0	68.2	67.0	62.8	56.7	49.5	42.7	
		MCDB	44.9	50.6	62.3	68.2	74.8	83.7	86.9	85.8	81.8	72.2	57.1	49.3	
5%		WB	37.3	40.6	46.5	53.0	58.9	63.6	66.8	65.9	61.3	55.3	47.6	40.5	
		MCDB	42.5	47.7	58.6	64.9	73.4	82.2	85.5	84.9	79.8	70.2	55.5	46.2	
10%		WB	35.4	38.6	44.4	51.3	57.4	62.3	65.6	64.7	59.9	53.9	45.8	38.6	
		MCDB	40.1	45.2	54.9	62.5	71.0	80.6	84.1	84.2	78.0	67.6	53.5	43.5	
Mean Daily Temperature Range		5% DB	MDBR	16.3	18.4	20.0	21.2	23.5	25.6	25.3	27.1	27.6	24.3	19.7	16.8
			MCDBR	20.6	22.9	25.3	26.2	27.2	29.0	30.1	31.3	31.7	29.2	24.3	21.8
	MCWBR		13.9	13.9	13.3	12.9	12.0	12.3	13.1	13.1	13.7	13.5	13.6	13.8	
	5% WB	MCDBR	18.6	20.8	24.0	22.9	25.0	26.6	26.1	27.5	28.8	26.5	21.1	19.4	
		MCWBR	13.0	13.4	13.3	12.0	12.0	12.4	12.1	12.3	13.4	13.2	12.5	12.8	
Clear-Sky Solar Irradiance	taub	0.302	0.335	0.385	0.452	0.443	0.411	0.442	0.414	0.370	0.382	0.326	0.301		
	taud	2.366	2.290	2.156	1.988	2.068	2.203	2.125	2.204	2.318	2.224	2.372	2.406		
	Ebn at Noon	279	282	276	263	268	276	266	271	278	260	267	271		
	Edh at Noon	29	35	44	56	52	46	49	44	37	37	28	26		
All-Sky Solar Radiation	RadAvg	794	1115	1373	1713	2102	2555	2481	2253	1850	1238	852	670		
	RadStd	65	104	109	144	122	126	77	90	106	97	104	83		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-3.22	N/A	+2.25	N/A	-0.68	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	+1.40	N/A	-0.90	N/A	N/A	N/A	N/A

Nomenclature: See separate page